MIL-M-45665 (Ord)
22 April 1960

SUPERSEDING
FA(FC)-PD-8
2 May 1958

MILITARY SPECIFICATION

MOUNT, TELESCOPE: 8228899

1. SCOPE

1.1 This specification covers one type of mount known as Mount, Telescope: 8228899 which is a part of Optical Sightunit: XM43 used with "Honest John" Rocket Launcher, XM33.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

Military

MIL-M-12192	Mounts, Telescope, M79, M79A1, and 8228899, Packaging of
MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-130	Identification Marking of U. S. Military Property

DRAWINGS

Ordnance Corps

F8228899	Mount, Telescope: 8228899

(Copies of specifications, standards, and drawings required by the contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 1240

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3. REQUIREMENTS

3.1 Production model.- At the beginning of production, one mount representative of production, shall be made available to the Government to determine if the mount meets all the requirements of this specification.

3.2 Materials.- Material shall be in accordance with applicable drawings and specifications forming a part of this specification.

3.3 Construction.- The mount shall be manufactured in accordance with Drawing F8228899, "Mount, Telescope: 8228899" and drawings pertaining thereto. The requirements of this specification are detailed only to the extent considered necessary to insure the inherent quality and the performance capabilities of the product prior to final acceptance by the Government.

3.4 General specification.- The contractor shall be responsible for adherence to, and compliance with, the following requirements of Specification MIL-F-13926:

(a) Order of precedence
(b) Protective finished surfaces
(c) Special accuracy
(d) Finish, painting and protective
(e) Workmanship

3.5 Environmental.

3.5.1 Operating temperature.- The mount shall be operable and shall meet the requirements specified in 3.6.7 and 3.6.8 while exposed to and thermally stabilized at ambient temperatures of plus 150 and minus 65 degrees Fahrenheit.

3.5.2 Vibration.- The mount shall meet all requirements of this specification after being vibrated in a vertical plane in the normal operating position for five minutes, plus or minus 15 seconds at an amplitude of 1/16 inch (1/8 inch total displacement) at a frequency of 30 cycles per second. During vibration the mount shall support an Elbow Telescope: M62A1D, or equivalent, in size and weight.

3.6 Performance.- All performance requirements shall be met with the mount positioned as follows:

(a) The dovetail horizontal within 2 minutes.
(b) The top locating surface of the mount which receives Adapter Telescope: M9A1, parallel to the dovetail locating surface within 1 minute.

3.6.1 Level vial.- The elevation level vial bubble shall be central within the width of a graduation line.

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3.6.2 Elevation scale.- The zero graduation of the elevation scale and its micrometer shall be in coincidence with its respective index.

3.6.3 Plumb travel.- Utilizing the mounting surface that locates the Adapter Telescope, M9A1, a line of sight shall be established perpendicular to the axis of the dovetail. The line of sight shall not deviate from a plumb line by more than plus or minus 0.5 mil (total deviation not to exceed 0.5 mil) when the elevation knob is rotated for the entire elevation range.

3.6.3.1 Side play.- When the line of sight is in coincidence with the plumb line, the line of sight shall not shift from the plumb line by more than 0.25 mil when the elevation knob rotation is reversed.

3.6.4 Elevation error.- The elevation mechanism error shall not exceed 1.0 mil excluding backlash at any reading of the elevation scale.

3.6.4.1 Backlash.- Backlash error in the elevating mechanism shall not exceed 0.5 mil.

3.6.5 Elevation level vial.- The elevation level bubble shall not deviate from the centered position by more than one half a vial division when the mount is canted 15 degrees each side of center by rotating the cross leveling knob.

3.6.6 Cant movement.- There shall be at least 30 degrees cant movement on each side of the horizontal position before contacting the cross level stops.

3.6.7 Dovetail clamp lever.- When subjected to the operating temperatures specified in 3.5.1, the dovetail clamp lever shall be capable of firmly securing the mount assembly to a dovetail plate (which mounts on the launcher) when actuated, and when released, the mount assembly shall be removable without force or restriction.

3.6.8 Torque.- The torque required to operate the knobs at standard ambient temperature (60 to 90 degrees Fahrenheit) shall be within the tolerance specified in Table I below. When measured at the extreme operating temperatures specified in 3.5.1, the torque values shall not have changed from the respective values obtained at standard ambient temperature by more than 3 pound inches.

TABLE I - TORQUE (pound-inches)

	Minimum	Maximum
Elevation knob	4	10
Cant correction knob	4	13

3.7 Identification marking.- Unless otherwise specified, idenfication marking shall be in accordance with Standard MIL-STD-130.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor inspection.- Unless otherwise specified herein, the supplier is responsible for the performance of all inspection requirements prior to submission for Government inspection and acceptance. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order.

4.2 Contractor's quality assurance system.- The contractor shall provide and maintain an adequate quality assurance system acceptable to the Government covering the supplies under the contract. A current written description of the system shall be submitted to the contracting officer prior to initiation of production. The written description will be considered acceptable when it provides:

(a) The quality assurance required by the applicable drawings and specifications listed in the contract or order.
(b) Contractor's quality standards of acceptability.
(c) Procedure for continually evaluating product quality.
(d) Procedure for continually reporting evidence of actual product quality.
(e) Measures for preventing any significant departure of product quality from standards.

4.2.1 The contractor will not be restricted to the inspection station or to the method of inspection listed provided that an equivalent control is included in the approved quality assurance procedure. In cases of dispute as to whether or not certain procedures of the system provide equal assurance, the comparable procedure required by this specification shall be used. The contractor shall notify the Government of and obtain approval for any change to the written procedure that might affect the degree of assurance required by this specification and other applicable documents referenced herein.

4.3 Government verification.- All quality assurance operations performed by the contractor will be subject to Government verification at unscheduled intervals. Verification will consist of (a) surveillance of the operations to determine that the practices, methods, and procedures of the written inspection plan are being properly applied, and (b) Government product inspection to measure quality of product offered for acceptance. Deviation from the prescribed or agreed-upon procedures, or instances of poor practices which might have a detrimental affect upon the quality of the product, will be immediately called to the attention of the contractor. Failure of the contractor to promptly correct deficiencies discovered will be cause for suspension of acceptance until correction has been made or until conformance of product to prescribed criteria has been demonstrated. To avoid interference with operations, the

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contractor will designate a responsible official or officials to whom the Government inspector will report such instances.

4.4 Production sample.- Unless otherwise specified in the contract, one mount shall be selected by the Government inspector as soon as practicable after the start of production, and be inspected by the contractor under the surveillance of the Government. This inspection shall be conducted at the contractor's facility, or Government approved laboratory, using test equipment authorized for use on the contract. The sample shall be inspected for all requirements of this specification by utilizing examination and testing methods prescribed herein.

4.4.1 Production sample failure.- Should the production sample fail to pass the prescribed examination and tests, the contractor shall cease those operations which are causing the rejection and shall make the necessary corrections. When corrections in the process have been made, a new production sample shall be examined and tested by the contractor under the surveillance of the Government. Acceptance of production lots will not commence until an acceptable production sample is submitted and corrective actions have been taken to eliminate deficiencies.

4.5 Inspection provisions.

4.5.1 Lot formation.- Unless otherwise specified by the contracting officer, or his representative, inspection lot sizes and lot formation shall be in accordance with Standard MIL-STD-105.

4.5.2 Inspection of components.- All positional and functional characteristics shown on the drawings contained in the contract shall be inspected by the contractor for conformance with drawing requirements prior to acceptance. These characteristics are defined as drawing requirements effecting concentricity, squareness, parallelism, micro-finishes 64 RMS or better, threads, identification marking, and all dimensions on which the tolerances indicate a precision requirement that have a bearing on the functional accuracy and performance of the mount. Also included in the above category are all performance requirements specified in notes directly on the drawings, or in separate performance specifications referenced therein.

4.5.2.1 Sampling inspection.- Sampling inspection of components may be used in liew of 100 percent inspection, and shall be in accordance with Table II.

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TABLE II - SAMPLING PLAN

Lot Size Inclusive	Sample Size	Accept Lot	Reject Lot
0 - 25	Use 100% Inspection		
26 - 35	20	0	1
36 - 50	25	0	1
51 - 150	30	0	1
151 - 300	30	1	2
301 - over	60	2	3

4.5.2.2 Disposition of non-conforming product.- Acceptance and rejection numbers are expressed in terms of defective units. One or more defects on any unit shall constitute a defective unit. Disposition of non-conforming components shall be in accordance with Standard MIL-STD-105.

4.6 Control of materials.- Materials shall be tested to determine conformance to contractually applicable specifications. Such tests may be conducted in the prime contractor's or subcontractor's facilities, or in other suitably equipped laboratories. The test results shall be available to the Government inspector for review. Certified test reports, identifiable with the material, may be accepted in lieu of repeated tests. The Government reserves the right, in any case, to require retests to assure that the materials are acceptable.

4.7 Acceptance tests.

4.7.1 Classification of defects.- The classification of defects in Tables III and IV shall constitute the minimum inspection requirements of telescope mount to be performed by the contractor prior to acceptance or rejection by lot. The Government reserves the right to inspect for any applicable requirement, and to reject individual non-conforming items.

4.7.2 Disposition of rejected lots.- Rejected lots shall be screened for all defective characteristics. After corrections have been made and the Government inspector informed of the corrective action taken, the lot shall be resubmitted for acceptance as outlined in Standard MIL-STD-105.

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TABLE III - CLASSIFICATION OF DEFECTS

Use Inspection Level L7, and Sampling Plan Table IV-A of Standard MIL-STD-105

MAJOR: AQL 4.0%		Req't Par.	QA Par.	Test Method
1.	Operating temperature	3.5.1	4.9.1	Device, Special Testing (DST)
2.	Vibration	3.5.2	4.9.2	DST

TABLE IV - CLASSIFICATION OF DEFECTS

Use Inspection Level II, and Sampling Plan Table IV-A of Standard MIL-STD-105

MAJOR: AQL 1.5%		Req't Par.	QA Par.	Test Method
1.	Performance	3.6	4.9.3.1	DST
2.	Level vial	3.6.1	4.9.3.2	Visual-DST
3.	Elevation scale	3.6.2	4.9.3.3	Visual-DST
4.	Plumb travel	3.6.3	4.9.3.4	DST
5.	Side play	3.6.3.1	4.9.3.5	DST
6.	Elevation error	3.6.4	4.9.3.6	DST
7.	Backlash	3.6.4.1	4.9.3.7	DST
8.	Elevation level vial	3.6.5	4.9.3.8	Visual-DST
9.	Cant movement	3.6.6	4.9.3.9	DST
10.	Torque	3.6.8	4.9.3.11	DST
MINOR: AQL 4.0%				
41.	Dovetail clamp lever	3.6.7	4.9.3.10	Manual-DST
42.	Identification marking	3.7	4.9.4	Visual

4.8 Inspection equipment.

4.8.1 Contractor provisions.- Unless otherwise specified in the contract, the contractor shall provide and properly utilize all gaging, measuring, and testing equipment of the required accuracy for the performance of all inspection examinations and tests required to be performed by the contractor under this contract. The contractor shall be responsible for the maintenance of the equipment including surveillance, calibration, and rehabilitation, on a regularly scheduled basis, and shall have available, all calibration equipment and other instrumentation essential for calibration and certification of the inspection equipment. The contractor also shall maintain records of inspection equipment surveillance and calibration, and shall make these records available for review by the Government. The Government reserves the right to certify inspection equipment, and to use it for Government verification inspection or acceptance inspection.

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4.8.1.1 Accuracy requirements.- Inspection equipment which incorporates features for measuring values or sizes, established by product requirements, shall be capable of measuring these values within an accuracy of ten percent of the specified tolerance. In cases where the ten percent accuracy imposes impractical or very difficult design and fabrication problems, or results in a test equipment cost factor which is exorbitant in relation to the cost and application of the product to be inspected, the Government shall be expeditiously advised in order that necessary appropriate action can be taken.

4.9 Test methods and procedures.

4.9.1 Operating temperature.- The test shall consist of exposing the mounts to the temperatures indicated in 3.5.1 by means of a cold chest and a heating chamber capable of maintaining the specified temperatures until the mounts under test are thermally stabilized. While at these temperatures, the mount shall be checked for compliance with the performance requirements specified in 3.5.1.

4.9.2 Vibration.- The mount shall be vibrated as indicated in 3.5.2 by means of a vibrating machine on which the mount shall be supported by a device for holding the mount in its normal operating position. The vibrating machine shall be capable of producing a vertical motion at the frequency, amplitude and duration specified. Upon completion of the vibration test, the mount shall be visually examined for evidence of damage, then checked to determine compliance with the requirements specified in 3.5.2.

4.9.3 Performance tests.- The performance tests shall be accomplished by means of a special testing device. The device shall provide the means for attaching and locating the dovetail surface in a horizontal plane and provide the means for plumb travel movement.

4.9.3.1 Performance tests set-up.- The set-up for performance tests shall be made by cross-leveling the special testing device by means of ten second bench level and plumb travel movement. The telescope mount shall be attached to the special testing device with the flat of the dovetail mounting surface in a horizontal plane. With the ten second bench level positioned across the top surface of the mount, the indication on the bench level shall not exceed the tolerance of 3.6(a). With the bench level positioned across the top locating surface of the mount which receives the telescope adapter, the indication on the bench level shall not exceed the tolerance of 3.6(b).

4.9.3.2 Level vial.- The level bubble shall be visually checked for compliance with the requirements of 3.6.1 when the mount is positioned as required in 4.9.3.1.

4.9.3.3 Elevation scale.- Subsequent to the above examination, a visual check shall be made to determine compliance with the requirements of 3.6.2.

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4.9.3.4 Plumb travel.- The test shall be performed with the mount attached to a special testing device as indicated in 4.9.3.1. A target for checking the plumb travel shall be a plumb line located approximately fifty feet from the test fixture and in alinement with an auxiliary telescope positioned in a special testing device secured to the mounting surface that locates the Adapter Telescope, M9A1. A mil block chart in back of the plumb line will be used to determine if the line-of-sight as established by the testing device will track plumb within the tolerance specified in 3.6.3. By means of the elevation knob, rotate the line-of-sight through its entire movement while simultaneously rotating the elevation test fixture in opposite directions. In either up or down movement the line-of-sight shall not deviate from the plumb line more than the tolerance specified. A peep sight with an approximate 1/16 inch aperture placed over the eyepiece of the auxiliary telescope will assist the observer's eye.

4.9.3.5 Side play.- The side play check shall be made as indicated in 3.6.3.1 in conjunction with a mil block chart located in back of the plumb line.

4.9.3.6 Elevation error.- The test shall be accomplished with the same set-up as used for the plumb travel test of 4.9.3.4. The elevation knob and the special testing device shall be set at zero when the auxiliary telescope is focused on the mil block target. The elevation knob shall be moved to a particular reading, and then the special testing device shall be rotated in the opposite direction until the telescope is in coincidence with the target. The reading taken on the scale and vernier of the device will indicate the error, if any. Indication of any error shall not exceed the tolerance of 3.6.4. The procedure shall be repeated, and unless otherwise specified, a minimum of five readings shall be taken at approximately equidistant points throughout the entire elevation movement.

4.9.3.7 Backlash.- Backlash of the elevating mechanism can be measured by means of the elevation knob and a wall target positioned as in 4.9.3.4. Backlash shall be checked at positions at which accuracy readings are taken. At each position checked, the mount shall be brought in coincidence with the target with a clockwise movement, and the reading noted. The clockwise movement shall be continued beyond the target and then brought in coincidence with the target with a counterclockwise movement, and reading noted. The difference between the readings shall be considered as backlash, and shall not exceed the tolerance of 3.6.4.1.

4.9.3.8 Elevation level vial.- With the mount positioned on the special testing device, and with the aid of an optical protractor, the mount shall be canted 15 degrees each side of center position by rotating the cross-leveling knob. The elevation level vial bubble shall be visually checked to determine compliance with the requirements specified in 3.6.5.

4.9.3.9 Cant movement.- The cant movement test shall be made with the same set-up as employed in the elevation level vial test of 4.9.3.8. The test shall consist of positioning the mount in the cant position and observing the optical protractor while rotating the cross-leveling knob throughout its entire

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excursion in either direction to determine compliance with the requirements of 3.6.6.

4.9.3.10 Dovetail clamp lever.- The dovetail clamp lever shall be manually tested for operability when subjected to the temperatures specified in 3.5.1. The special testing device indicated in 4.9.3.1 will be used for this test to determine compliance with the requirements of 3.6.7.

4.9.3.11 Torque.- The torque required to operate the knobs of the mount through full travel shall be made by means of a commercial torque wrench with a special adapter. The measured torque shall not exceed the tolerance specified in 3.6.8 for the temperatures indicated.

4.9.4 Identification marking.- Marking for identification shall be visually inspected for conformance in accordance with Standard MIL-STD-130.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking.- Packaging, packing and marking shall be in accordance with Specification MIL-M-12192.

6. NOTES

6.1 Intended use.- The mount supports the Adapter Telescope: M9A1 and Elbow Telescope: M62A1D and incorporates a manual cant correcting and elevating mechanism for indirect fire.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Selection of applicable levels of preservation, packaging, and packing.
(c) Applicable stock number.
(d) Provisions for production model including date of delivery (see 3.1).

6.3 Design information.- Design information regarding specific equipment not covered or referenced by this specification may be obtained from the procuring activity upon the specific request of the contractor.

6.4 Definitions.

6.4.1 Device, special testing (DST).- Special testing devices are defined as special inspection fixtures, instruments, and modified standard measuring equipment.

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procurement operation, the United States Government thereby incurs no responsibility nor any obligations whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any invention that may in any way be related thereto.

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